

Clinical Concepts & Medical Glossary for Engineers

ENGINEER'S MEDICAL PRIMER

Software engineers working on clinical decision support systems do not need a medical degree. This primer translates nephrology, renal hemodynamics, acute kidney injury (AKI), drug interactions, and laboratory panic values into intuitive mechanical, plumbing, and distributed system concepts.

1. The Kidney as a Water Filter (Creatinine & AKI)

The human kidneys are two continuous biological filtration units filtering waste out of 5 liters of circulating blood. Muscles steadily generate a harmless byproduct called **Creatinine**. When kidneys filter cleanly, blood creatinine stays low (0.7 - 1.1 mg/dL). If the filter gets clogged or damaged, creatinine builds up rapidly in the blood.

KDIGO Stage	Mathematical Trigger	Clinical Meaning & Physician Action
Stage 1 (Moderate)	Delta ≥ 0.3 mg/dL within 48h or 1.5x - 1.9x baseline	Mild acute filtration drop. Rehydrate, review medications, recheck in 24h.
Stage 2 (High Priority)	Creatinine 2.0x - 2.9x baseline (e.g. 1.0 $\rightarrow$ 2.4 mg/dL)	Significant injury. Discontinue offending nephrotoxic medications immediately.
Stage 3 (Critical)	Creatinine ≥ 3.0 x baseline or ≥ 4.0 mg/dL with acute rise	Near total filter shutdown. Emergency nephrology consult; possible dialysis.

2. Renal Hemodynamics & The 'Triple Whammy'

To filter waste, the kidney requires high internal water pressure. Picture a kitchen sink with an inlet faucet and an outlet drain pipe:

- **NSAIDs (Advil / Ibuprofen / Naproxen):** Constrict the inlet faucet, cutting off incoming blood volume.
- **ACE Inhibitors & ARBs (Lisinopril / Losartan):** Dilate (open wide) the drain pipe, collapsing pressure.
- **Diuretics (Furosemide / Water Pills):** Drain total circulating body fluid.
- **The Triple Whammy (Contraindicated):** Taking all three together causes internal kidney pressure to collapse to near zero, precipitating acute prerenal kidney failure.

3. Electrolytes: The Heart's Electrical Grid

Biomarker	Normal Range	Panic Threshold	Life-Threatening Mechanism
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Potassium (K+)	3.5 - 5.0 mEq/L	> 6.0 mEq/L (Hyperkalemia)	Disrupts cardiac electrical resting potential; causes ventricular fibrillation / cardiac arrest.
Hemoglobin (Hb)	12.0 - 17.5 g/dL	< 7.0 g/dL (Severe Anemia)	Blood cannot carry adequate oxygen to brain and organs; requires urgent blood transfusion.

4. Healthcare Ontologies & Code Standards

Standard	Domain Governed	Codebase Example	Plain English Meaning
FHIR R4	JSON API Graph Schema	resourceType: Observation	Standard interoperable health record payload.
LOINC	Laboratory Tests	2160-0	Serum Creatinine concentration test.
RxNorm	Medications & Dosages	314076	Lisinopril 10 MG Oral Tablet.
SNOMED-CT	Diagnoses & Diseases	38341003	Essential Hypertension (high blood pressure).

KEY CODEBASE CONSTANTS & MEANINGS

CODE-TO-CLINIC ROSETTA STONE:

- *KDIGO_AKI_STAGE_2 = Patient's kidney filter is 50% failing; hold blood pressure meds & NSAIDs.*
- *TRIPLE_WHAMMY = Blood pressure pill + water pill + Advil taking together; stop Advil immediately.*
- *PANIC_VALUE (K+ > 6.0) = Potassium spike risking cardiac arrest; order emergent stat ECG.*
- *triage_level: HIGH = Critical bedside emergency; move patient to top of rounding queue.*
- *triage_level: LOW = Stable baseline vitals; safe for standard outpatient observation.*